

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('/content/anonymized_marks.csv') # change name if needed
df.head()
```

	Student	Marks	
0	Student1	14.5	
1	Student2	1.0	
2	Student3	5.0	
3	Student4	5.0	
4	Student5	13.0	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
print(df.columns)

Index(['Student', 'Marks'], dtype='object')
```

```
col = "Marks" # change if needed

data = df[col].dropna()

print(" ♦ Highest (Max):", data.max())
print(" ♦ Lowest (Min):", data.min())

print(" ♦ Mean:", data.mean())
print(" ♦ Median:", data.median())
print(" ♦ Mode:", data.mode().tolist())

print(" ♦ Standard Deviation:", data.std())
print(" ♦ Variance:", data.var())
```

- ♦ Highest (Max): 17.5
- ♦ Lowest (Min): 0.0
- ♦ Mean: 7.88135593220339
- ♦ Median: 6.0
- ♦ Mode: [6.0, 11.5]
- ♦ Standard Deviation: 5.03294232776179
- ♦ Variance: 25.33050847457627

```
stats = pd.DataFrame({
    "Min": [data.min()],
    "Max": [data.max()],
    "Mean": [data.mean()],
    "Median": [data.median()],
    "Mode": [data.mode().tolist()],
    "Std Dev": [data.std()],
    "Variance": [data.var()]
})

stats
```

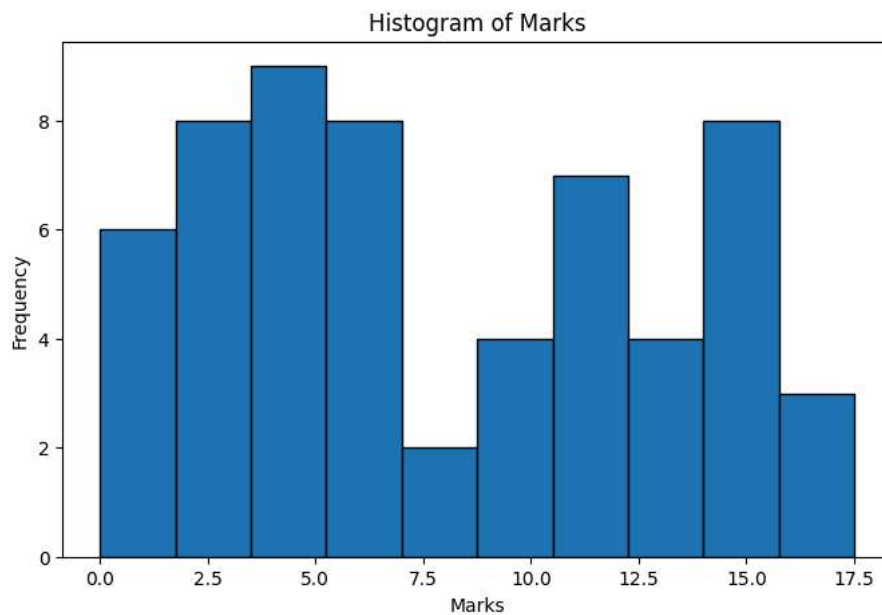
	Min	Max	Mean	Median	Mode	Std Dev	Variance	
0	0.0	17.5	7.881356	6.0	[6.0, 11.5]	5.032942	25.330508	

```
df.describe()
```

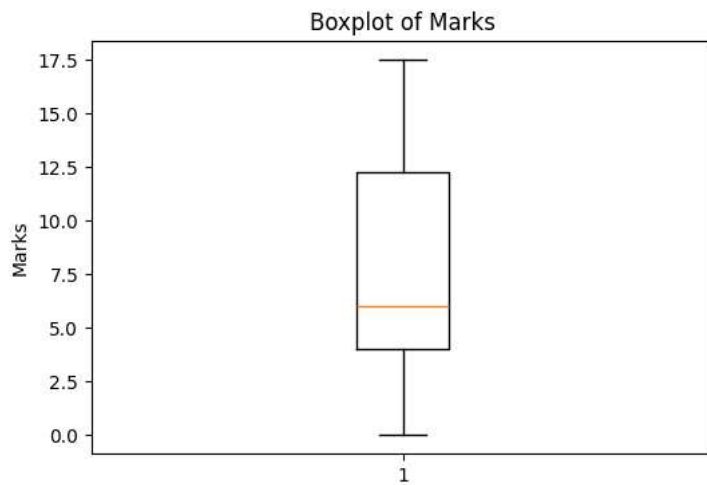
	Marks
count	59.000000
mean	7.881356
std	5.032942
min	0.000000
25%	4.000000
50%	6.000000
75%	12.250000
max	17.500000

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(data, bins=10, edgecolor='black')
plt.title("Histogram of Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```



```
plt.figure(figsize=(6,4))
plt.boxplot(data, vert=True)
plt.title("Boxplot of Marks")
plt.ylabel("Marks")
plt.show()
```



```
print("Skewness:", data.skew())
print("Kurtosis:", data.kurt())
```

```
Skewness: 0.22573000497649964
Kurtosis: -1.2984354913969534
```

```
Q1 = data.quantile(0.25)
Q2 = data.quantile(0.50)
Q3 = data.quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = data[(data < lower_bound) | (data > upper_bound)]

print("Q1:", Q1)
print("Q2 (Median):", Q2)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:")
print(outliers.tolist())
```

```
Q1: 4.0
Q2 (Median): 6.0
Q3: 12.25
IQR: 8.25
Lower Bound: -8.375
Upper Bound: 24.625
Outliers:
[]
```

```
summary = pd.DataFrame({
    "Minimum": [data.min()],
    "Maximum": [data.max()],
    "Range": [data.max() - data.min()],
    "Mean": [data.mean()],
    "Median": [data.median()],
    "Mode": [data.mode().tolist()],
    "Standard Deviation": [data.std()],
    "Variance": [data.var()],
    "Q1": [Q1],
    "Q3": [Q3],
    "IQR": [IQR],
    "Skewness": [data.skew()],
    "Kurtosis": [data.kurt()],
    "Number of Outliers": [len(outliers)]
})

summary
```

	Minimum	Maximum	Range	Mean	Median	Mode	Standard Deviation	Variance	Q1	Q3	IQR	Skewness	Kurtosis	Number of Outliers
0	0.0	17.5	17.5	7.881356	6.0	[6.0, 11.5]	5.032942	25.330508	4.0	12.25	8.25	0.22573	-1.298435	0

```

print("INTERPRETATION:")
print(f"1. Highest marks = {data.max()} and lowest marks = {data.min()}.")
print(f"2. Average marks (mean) = {data.mean():.2f}.")
print(f"3. Median marks = {data.median():.2f}.")
print(f"4. Mode = {data.mode().tolist()}")
print(f"5. Standard deviation = {data.std():.2f}, showing the spread of marks.")
print(f"6. Variance = {data.var():.2f}.")
print(f"7. Skewness = {data.skew():.2f}.")
print(f"8. Kurtosis = {data.kurt():.2f}.")
print(f"9. Number of outliers = {len(outliers)}")

```